ER Diagram for Municipal OTMS (Online Tax Management System)

Core Entities

1. Ward

Represents municipal administrative divisions.

Attributes:

- ward_id (PK)
- ward_number
- ward_name
- zone_name
- officer_name
- contact_number

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2. Property

Represents taxable properties within a ward.

Attributes:

- property_id (PK)
- ward_id (FK)
- property_number
- property_type
- usage_type
- address
- area_sqft
- construction_year
- occupancy_status
- latitude
- longitude
- status

---

3. Owner

Represents owners of properties.

Attributes:

- owner_id (PK)
- full_name
- father_name
- mobile_number
- email
- aadhaar_number
- address
- gender
- date_of_birth

---

4. Property_Owner

Handles multiple owners for one property.

Attributes:

- property_owner_id (PK)
- property_id (FK)
- owner_id (FK)
- ownership_percentage
- ownership_type
- start_date
- end_date

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5. Tax_Assessment

Stores yearly tax assessments including house tax, water tax, and sewer tax.

Attributes:

- assessment_id (PK)
- property_id (FK)
- financial_year
- annual_rental_value
- house_tax_amount
- water_tax_amount
- sewer_tax_amount
- other_charges
- tax_rate
- rebate_amount
- penalty_amount
- total_tax

- assessment_date
- assessed_by

---

6. Tax_Payment

Stores payment records.

Attributes:

- payment_id (PK)
- assessment_id (FK)
- payment_date
- payment_mode
- transaction_reference
- paid_amount
- receipt_number
- payment_status

---

7. Tax_Rate_Master

Stores tax rates by property category.

Attributes:

- tax_rate_id (PK)
- property_type
- usage_type
- tax_percentage
- effective_from
- effective_to

---

8. Employee

Municipal staff handling operations.

Attributes:

- employee_id (PK)
- employee_name
- designation
- mobile_number
- email
- ward_id (FK)

- username
- password_hash
- role

---

9. Mutation

Handles ownership transfer and property mutation records.

Attributes:

- mutation_id (PK)
- property_id (FK)
- old_owner_id (FK)
- new_owner_id (FK)
- mutation_date
- mutation_type
- approval_status
- approved_by
- remarks
- document_reference

Relationships

Entity 1	Relationship	Entity 2
Ward	One-to-Many	Property
Property	Many-to-Many	Owner
Property	One-to-Many	Tax_Assessment
Tax_Assessment	One-to-Many	Tax_Payment
Tax_Rate_Master	One-to-Many	Tax_Assessment
Ward	One-to-Many	Employee
Property	One-to-Many	Mutation
Owner	One-to-Many	Mutation (Old Owner)
Owner	One-to-Many	Mutation (New Owner)

ER Diagram (Mermaid Syntax)

```
erDiagram
    WARD ||--o{ PROPERTY : contains
    PROPERTY ||--o{ PROPERTY_OWNER : mapped
    OWNER ||--o{ PROPERTY_OWNER : owns
    PROPERTY ||--o{ TAX_ASSESSMENT : assessed_for
    TAX_ASSESSMENT ||--o{ TAX_PAYMENT : paid_by
    TAX_RATE_MASTER ||--o{ TAX_ASSESSMENT : applies
    WARD ||--o{ EMPLOYEE : managed_by
    PROPERTY ||--o{ MUTATION : mutation_for
    OWNER ||--o{ MUTATION : old_owner
    OWNER ||--o{ MUTATION : new_owner

    WARD {
        int ward_id PK
        string ward_number
        string ward_name
        string zone_name
    }

    PROPERTY {
        int property_id PK
        int ward_id FK
        string property_number
        string property_type
        string usage_type
        string address
        float area_sqft
    }

    OWNER {
        int owner_id PK
        string full_name
        string mobile_number
        string aadhaar_number
    }

    PROPERTY_OWNER {
        int property_owner_id PK
        int property_id FK
        int owner_id FK
        float ownership_percentage
    }

    TAX_ASSESSMENT {
```

```
    int assessment_id PK
    int property_id FK
    string financial_year
    decimal house_tax_amount
    decimal water_tax_amount
    decimal sewer_tax_amount
    decimal other_charges
    decimal total_tax
    decimal rebate_amount
    decimal penalty_amount
}

TAX_PAYMENT {
    int payment_id PK
    int assessment_id FK
    date payment_date
    decimal paid_amount
    string payment_mode
}

TAX_RATE_MASTER {
    int tax_rate_id PK
    string property_type
    string usage_type
    decimal tax_percentage
}

EMPLOYEE {
    int employee_id PK
    string employee_name
    string designation
    int ward_id FK
}

MUTATION {
    int mutation_id PK
    int property_id FK
    int old_owner_id FK
    int new_owner_id FK
    date mutation_date
    string mutation_type
    string approval_status
}
```

Suggested Additional Modules

You can also extend the OTMS with:

- Advanced Mutation Workflow
- Water Tax
- Sewer Tax
- GIS Mapping
- Complaint Management
- Notice Generation
- Online Payment Gateway
- Audit Logs
- User Roles & Permissions
- Mobile App Integration

SQL Schema (MySQL/PostgreSQL Compatible)

```
CREATE TABLE Ward (
    ward_id INT PRIMARY KEY AUTO_INCREMENT,
    ward_number VARCHAR(20) NOT NULL,
    ward_name VARCHAR(100),
    zone_name VARCHAR(100),
    officer_name VARCHAR(100),
    contact_number VARCHAR(15)
);

CREATE TABLE Property (
    property_id INT PRIMARY KEY AUTO_INCREMENT,
    ward_id INT,
    property_number VARCHAR(50) UNIQUE NOT NULL,
    property_type VARCHAR(50),
    usage_type VARCHAR(50),
    address TEXT,
    area_sqft DECIMAL(10,2),
    construction_year INT,
    occupancy_status VARCHAR(50),
    latitude DECIMAL(10,6),
    longitude DECIMAL(10,6),
    status VARCHAR(20),
    FOREIGN KEY (ward_id) REFERENCES Ward(ward_id)
);

CREATE TABLE Owner (
    owner_id INT PRIMARY KEY AUTO_INCREMENT,
```

```sql
    full_name VARCHAR(100) NOT NULL,
    father_name VARCHAR(100),
    mobile_number VARCHAR(15),
    email VARCHAR(100),
    aadhaar_number VARCHAR(20),
    address TEXT,
    gender VARCHAR(10),
    date_of_birth DATE
);

CREATE TABLE Property_Owner (
    property_owner_id INT PRIMARY KEY AUTO_INCREMENT,
    property_id INT,
    owner_id INT,
    ownership_percentage DECIMAL(5,2),
    ownership_type VARCHAR(50),
    start_date DATE,
    end_date DATE,
    FOREIGN KEY (property_id) REFERENCES Property(property_id),
    FOREIGN KEY (owner_id) REFERENCES Owner(owner_id)
);

CREATE TABLE Tax_Rate_Master (
    tax_rate_id INT PRIMARY KEY AUTO_INCREMENT,
    property_type VARCHAR(50),
    usage_type VARCHAR(50),
    tax_percentage DECIMAL(5,2),
    effective_from DATE,
    effective_to DATE
);

CREATE TABLE Tax_Assessment (
    assessment_id INT PRIMARY KEY AUTO_INCREMENT,
    property_id INT,
    financial_year VARCHAR(20),
    annual_rental_value DECIMAL(12,2),
    house_tax_amount DECIMAL(12,2),
    water_tax_amount DECIMAL(12,2),
    sewer_tax_amount DECIMAL(12,2),
    other_charges DECIMAL(12,2),
    tax_rate DECIMAL(5,2),
    rebate_amount DECIMAL(12,2),
    penalty_amount DECIMAL(12,2),
    total_tax DECIMAL(12,2),
    assessment_date DATE,
    assessed_by VARCHAR(100),
    FOREIGN KEY (property_id) REFERENCES Property(property_id)
);
```

```sql
CREATE TABLE Tax_Payment (
    payment_id INT PRIMARY KEY AUTO_INCREMENT,
    assessment_id INT,
    payment_date DATE,
    payment_mode VARCHAR(50),
    transaction_reference VARCHAR(100),
    paid_amount DECIMAL(12,2),
    receipt_number VARCHAR(100),
    payment_status VARCHAR(20),
    FOREIGN KEY (assessment_id) REFERENCES Tax_Assessment(assessment_id)
);

CREATE TABLE Employee (
    employee_id INT PRIMARY KEY AUTO_INCREMENT,
    employee_name VARCHAR(100),
    designation VARCHAR(100),
    mobile_number VARCHAR(15),
    email VARCHAR(100),
    ward_id INT,
    username VARCHAR(50),
    password_hash VARCHAR(255),
    role VARCHAR(50),
    FOREIGN KEY (ward_id) REFERENCES Ward(ward_id)
);

CREATE TABLE Mutation (
    mutation_id INT PRIMARY KEY AUTO_INCREMENT,
    property_id INT,
    old_owner_id INT,
    new_owner_id INT,
    mutation_date DATE,
    mutation_type VARCHAR(50),
    approval_status VARCHAR(30),
    approved_by VARCHAR(100),
    remarks TEXT,
    document_reference VARCHAR(255),
    FOREIGN KEY (property_id) REFERENCES Property(property_id),
    FOREIGN KEY (old_owner_id) REFERENCES Owner(owner_id),
    FOREIGN KEY (new_owner_id) REFERENCES Owner(owner_id)
);
```

Sample Tax Calculation Query

```sql
SELECT
    p.property_number,
    o.full_name AS owner_name,
    t.financial_year,
    t.house_tax_amount,
    t.water_tax_amount,
    t.sewer_tax_amount,
    t.other_charges,
    t.rebate_amount,
    t.penalty_amount,
    (
        t.house_tax_amount +
        t.water_tax_amount +
        t.sewer_tax_amount +
        t.other_charges -
        t.rebate_amount +
        t.penalty_amount
    ) AS final_tax
FROM Tax_Assessment t
JOIN Property p ON t.property_id = p.property_id
JOIN Property_Owner po ON p.property_id = po.property_id
JOIN Owner o ON po.owner_id = o.owner_id;
```

Recommended Database

- MySQL
- PostgreSQL
- SQL Server

PostgreSQL is recommended if GIS/location support is needed.